

1. Administrative & Study Identification

Full Study Title: Study to Evaluate the Efficacy, Immunogenicity, and Safety of RSVpreF in Adults. **Official Title:** A PHASE 3 STUDY TO EVALUATE THE EFFICACY, IMMUNOGENICITY, AND SAFETY OF RESPIRATORY SYNCYTIAL VIRUS (RSV) PREFUSION F SUBUNIT VACCINE IN ADULTS **Short Title/Acronym:** RENOIR (RSV vaccine Efficacy study in older adults Immunized against RSV disease) **Protocol Number:** C3671013 (EudraCT: 2021-003693-31) **NCT Number:** NCT05035212 **Sponsor Name:** Pfizer **Investigational Product:** RSVpreF / Abrysvo (Bivalent RSV prefusion F subunit vaccine) **Phase of Development:** Phase 3 **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy and safety of a single dose of RSVpreF in the prevention of RSV-associated lower respiratory tract disease (LRTI-RSV) defined by ≥ 2 and ≥ 3 symptoms in adults 60 years of age and older. **Secondary Objectives:** To evaluate the vaccine efficacy against severe RSV-LRTD and to measure the immunogenicity (neutralizing titers for RSV A and RSV B) of the RSVpreF vaccine 1 month post-vaccination and across multiple RSV seasons.

2.2 Background & Rationale Respiratory syncytial virus (RSV) is a contagious virus that causes serious respiratory illness, particularly in older adults. In the US alone, RSV accounts for approximately 177,000 hospitalizations and 14,000 deaths annually among older adults. This study was initiated because, prior to Abrysvo, there were no prophylactic options for this vulnerable population. The vaccine utilizes a stabilized prefusion F protein based on foundational NIH discoveries, which demonstrated that antibodies targeting this conformation are highly effective at blocking viral infection.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1 **Trial Status:** ACTIVE_NOT_RECRUITING

3.2 Planned Enrollment & Duration Target N: 38,861 participants **Number of Sites:** Conducted globally (including the United States, Canada, Netherlands, Finland, Argentina, Japan, and South Africa) **Estimated Study Start:** 31-Aug-2021 **Estimated Primary Completion:** Jan-2027 (Long-term follow up and sub-studies)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants ≥ 60 years of age. 2. Healthy participants or those with preexisting stable disease (defined as disease not requiring significant changes in therapy or hospitalization for worsening disease during the 6 weeks before enrollment). Specific criteria apply for stable HIV, HCV, or HBV. 3. Adults who are ambulatory and live in the community, or in assisted living/long-term care facilities that provide minimal assistance (primarily responsible for self-care). 4. Willing and able to comply with all scheduled visits, frequent symptom assessment by mobile device application, and self-collection of nasal swabs.

4.2 Exclusion Criteria 1. Bleeding diathesis or condition associated with prolonged bleeding that would contraindicate intramuscular injection. 2. History of severe adverse reaction associated with a vaccine and/or severe allergic reaction (e.g., anaphylaxis) to any component of the study intervention. 3. Serious chronic disorder including metastatic malignancy, end-stage renal disease with or without dialysis, or clinically unstable cardiac disease. 4. Immunocompromised individuals with known or suspected immunodeficiency. 5. Individuals receiving chronic systemic treatment with immunosuppressive therapy (e.g., cytotoxic agents, monoclonal antibodies, systemic corticosteroids, radiotherapy) from 60 days before study intervention. 6. Receipt of blood/plasma products or immunoglobulin within 60 days before study intervention. 7. Previous vaccination with any licensed or investigational RSV vaccine. 8. Prior history of any subtype of Guillain-Barré syndrome of any etiology (specifically noted for Substudy C).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency (Efficacy Study): A single 120 μg dose of RSVpreF (composed of equal 60 μg amounts of recombinant RSV prefusion F from subgroups A and B).
Route of Administration: Intramuscular (IM) injection. **Substudies (Revaccination Evaluation):** * **Substudy A:** Evaluates safety and immunogenicity of a second 120 μg dose of RSVpreF administered after a dosing interval of approximately 2 years. * **Substudy B:** Evaluates safety and immunogenicity of a second 120 μg dose of RSVpreF administered after a dosing interval of approximately 1 year. * **Substudy C:** Evaluates safety and immunogenicity of a second 120 μg dose of RSVpreF administered after a dosing interval of either 3 or 4 years.
Duration of Treatment: Single administration in the main efficacy study, followed by continuous efficacy surveillance for up to 24 months.

5.2 Concomitant & Prohibited Therapy Permitted: Standard ongoing therapies for stable preexisting conditions; inhaled/nebulized, intra-articular, or topical corticosteroids; short-term systemic corticosteroids (<14 days) if discontinued ≥ 28 days prior. **Prohibited:** Other investigational RSV vaccines, chronic systemic immunosuppressive therapy, or blood/plasma products within 60 days prior to vaccination.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening / Baseline	Day 0	Informed Consent, Randomization, Pre-vaccination blood draw, First Dose (Vaccination)
Interim 1	Month 1	Safety assessment, Post-vaccination immunogenicity blood draw
Interim 2	Months 6 & 12	Surveillance for acute respiratory illness (ARI), Safety Labs, Immunogenicity assessment
End of Study	Month 24	Final safety follow-up and completion of Season 2 efficacy surveillance

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- Vaccine Efficacy (VE) against first-episode RSV-associated lower respiratory tract illness (LRTI-RSV) in the first RSV season:** RT-PCR-confirmed RSV A and/or RSV B-associated acute respiratory illness (ARI) with ≥ 2 or ≥ 3 lower respiratory signs/symptoms lasting more than 1 day, with RT-PCR-confirmed RSV infection within 7 days of onset.
- Local Reactions:** Proportion of participants reporting prompted local reactions (pain at injection site, redness, swelling) within 7-days after vaccination.
- Systemic Events:** Proportion of participants reporting prompted systemic events (fever, fatigue/tiredness, headache, nausea, muscle pain, joint pain, vomiting, diarrhea) within 7-days after vaccination.

7.2 Secondary Endpoints

- Number of first episode of RSV-associated severe LRTI (sLRTI-RSV) in the first RSV season.
- Number of first episode of LRTI-RSV in the second RSV season.
- Number of first episode of LRTI-RSV across 2 RSV seasons.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collection of prompted local and systemic reactions within 7 days post-vaccination, alongside frequent respiratory symptom assessments utilizing a mobile device e-diary application. **Serious Adverse Events (SAEs):** Continuously monitored throughout the study follow-up duration (up to 24 months). **Data Monitoring Committee (DMC):** An independent, external Data Monitoring Committee was established to routinely assess interim efficacy and safety data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Evaluable Efficacy Population (participants complying with key protocol criteria, with surveillance starting on Day 15 post-vaccination); Safety Population. **Sample Size Justification:** The trial targeted an enrollment of 38,861 subjects (1:1 randomization) to ensure sufficient power to demonstrate efficacy, pre-specified as the lower bound of the confidence interval being >20%. **Handling of Missing Data:** Participants who did not experience an event and were not lost to follow-up were censored at the end of their respective surveillance periods.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine remote and onsite site monitoring to verify electronic participant diaries/mobile application data and proper handling of nasal swab samples. **Audit Trail:** Validated tracking for eCRF locking, electronic patient-reported outcomes (ePRO) for symptom diaries, and centralized laboratory assessments for PCR and viral neutralizing titers.

11. Study Identifiers & Governance

Primary ID: {{C3671013}} **Registry Number:** {{NCT05035212}}
Sponsor/Lead Organization: {{Pfizer}} **Study Title:** {{Study to Evaluate the Efficacy, Immunogenicity, and Safety of RSVpreF in Adults.}} **Official Regulatory Title:** {{A PHASE 3 STUDY TO EVALUATE THE EFFICACY, IMMUNOGENICITY, AND SAFETY OF RESPIRATORY SYNCYTIAL VIRUS (RSV) PREFUSION F SUBUNIT VACCINE IN ADULTS}}

12. Clinical Framework (RSV Specific)

Primary Condition: {{Lower Respiratory Tract Illness / Respiratory Syncytial Virus Infection}} **Diagnosis Threshold:** {{RT-PCR-confirmed RSV infection with ≥ 2 or ≥ 3 lower respiratory tract symptoms}} **Medical Methodology:** {{Prophylactic Vaccination}} **Optimization Target:** {{RSV-A and RSV-B neutralizing antibody response and clinical protection across multiple seasons}}

13. Study Procedures & Methodology

Management Pathway: {{Symptom-driven surveillance and RT-PCR testing}} **Education Protocol (HCP):** {{Site training on vaccine

administration, trial procedures, and AE tracking}} **Education Protocol (Patient):** {{Training on recognizing acute respiratory symptoms, self-collecting nasal swabs, and using the electronic mobile diary application}} **Quality Audit Mechanism:** {{External DMC review and central laboratory validation of RSV PCR and neutralizing assays}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to approximately 24 months for the main efficacy study (across 2 RSV seasons)}} **Onsite Visit Cadence:** {{Day 0, Month 1, Month 6, Month 12, Month 24}} **Remote Monitoring Frequency:** {{Frequent symptom assessment via mobile device application}} **Checklist Review Interval:** {{Continuous tracking during RSV seasons}}

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					